HW4 Report (Chapters 4 & 5): Subqueries and Table Expressions

CSCI331 – PairProgrammingGroups (PPG_6)

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1 Project Overview

This homework (HW4) focuses on developing and validating SQL solutions using:

- Chapter 4: **Subqueries**
- Chapter 5: **Table Expressions** (derived tables, CTEs, views, APPLY)

Per instructions, the **TSQLV6** database was used for testing correctness, and the final working queries were implemented and executed in **Northwinds2024Student**. The work was completed under the group category **PairProgrammingGroups** with group name **PPG_6**.

2 Presentation Walkthrough Structure (MP4)

The MP4 walkthrough followed this structure for a non-technical audience:

1. Intro (PPG_6 + HW4)
2. Proposition (Northwinds)
3. To-Do list and Gantt chart (planning proof)
4. Demo: TSQLV6 testing (short)
5. Demo: Northwinds final queries (main)
6. FY quarter output (final result table)
7. NACE Competencies + LLM statement
8. GitHub repository page (proof of deliverables)

3 Intro Script (Used in MP4)

> Hello everyone. This is HW4 for CSCI331, Chapters 4 and 5, for Pair Programming Group PPG_6. We developed SQL solutions using subqueries and table expressions. We used the TSQLV6 database only for testing and debugging our queries. Then we implemented and executed the final queries in the Northwinds2024Student database, because that is the required submission database. In this video, we show our planning using the To-Do list and Gantt chart, then demonstrate the SQL queries running, including the fiscal year quarter analysis.

4 Problem Proposition (Northwinds)

Final Proposition:
Northwinds needs visibility into order activity and shipping cost trends over time. The business must understand how many orders are being placed and how freight (shipping cost) changes across time periods in order to plan operations and control shipping expenses. Using the Northwinds2024Student database, this project produces SQL reports using subqueries and table expressions, and includes a fiscal year quarter summary that reports total orders and total freight by federal FY quarter from newest to oldest.

5 Planning Evidence: To-Do List and Gantt Chart

To-Do List: A task checklist was used to assign work and track completion (status: Not Started / In Progress / Done).
Gantt Chart: A timeline plan was created to schedule tasks between the homework availability date and deadline.

These planning artifacts were shown in the MP4 to demonstrate project management and task responsibility.

6 Demo: TSQLV6 Testing (Included all practice)

All Chapter 4 and Chapter 5 queries were first tested in **TSQLV6** to ensure correctness, including syntax validation and reasonable output. This step reduced risk before migrating the queries to Northwinds.

7 Demo: Northwinds Final Queries (Main)

After confirming logic in TSQLV6, the queries were modified and executed in **Northwinds2024Student**. The student Northwinds schema used tables such as:

- `Sales.[Order]`, `Sales.OrderDetail`, `Sales.Customer`
- `HumanResources.Employee`
- `Production.Product`, `Production.Supplier`

Some column names differed from typical Northwind naming, so the final scripts were adjusted to match the actual database structure.

8 FY Quarter Scalar Function and Output (Final)

A scalar function was created to compute Federal Fiscal Year quarters (FY starts Oct 1 and ends Sep 30). The analysis summarized:

- Total number of orders per FY quarter
- Total freight per FY quarter
- Sorted newest to oldest FY quarters

Table 1: Total Orders and Total Freight by Federal FY Quarter (Newest to Oldest)

FY_Quarter	TotalOrders	TotalFreight
FY2022-Q3	88	7078.65
FY2022-Q2	182	15115.40
FY2022-Q1	120	9712.34
FY2021-Q4	103	8770.40
FY2021-Q3	93	8253.15
FY2021-Q2	92	5729.24
FY2021-Q1	82	6471.04
FY2020-Q4	70	3808.83

FY Quarter Results (Output show after execution)

9 Planning Evidence: To-Do List and Gantt Chart

To-Do List (Screenshot)

Task I	Task	Deliverable/Proof	Owner	Start Dat	Due Date	Status	Notes / Link
T1	Restore databases in SSMS (TSQLV6 + Northwinds2024Student)	Screenshot showing both databases under Object Explorer > Databases	Amrina Qayyum	2026-03-02	2026-03-02	Done	Completed; ensure Northwinds has Sales.[Order] etc.
T2	Run Chapter 4 (Subqueries) exercises in TSQLV6	Working queries saved in TSQLV6 SQL file + screenshots of key outputs/errors fixed	Amrina Qayyum	2026-03-02	2026-03-02	Done	Used Sales.Orders / Sales.OrderDetails etc. as needed.
T3	Run Chapter 5 (Table Expressions) exercises in TSQLV6	Working queries saved in TSQLV6 SQL file + screenshots of key outputs	Amrina Qayyum	2026-03-02	2026-03-02	Done	CTE + derived table + view/function patterns tested.
T4	Create FY Quarter scalar function + report in TSQLV6	Function script + output table (FY_Quarter, TotalOrders, TotalFreight) screenshot	Amrina Qayyum	2026-03-02	2026-03-02	Done	Federal FY starts Oct 1; order by newest->oldest.
T5	Map Northwinds2024Student schema/tables/columns	List of real table names + screenshot (HumanResources.Employee, Sales.Customer, Sales.[Order], Sales.OrderDetail, ...)	Amrina Qayyum	2026-03-02	2026-03-02	Done	Confirmed: Sales.[Order], Sales.Customer, HumanResources.Employee, Sales.OrderDetail.
T6	Convert all Chapter 4+5 queries to Northwinds2024Student	Northwinds SQL file runs end-to-end without errors	Amrina Qayyum	2026-03-02	2026-03-03	Done	Used dynamic column detection where column names differ (Employee/Customer names/country).
T7	Create FY Quarter scalar function + report in Northwinds2024Student	FY output table screenshot (newest->oldest) + function script	Amrina Qayyum	2026-03-03	2026-03-03	Done	Used Sales.[Order] (OrderDate, Freight).
T8	Write Problem Proposition (Northwinds only)	1-paragraph proposition saved in README or report + shown in video	Amrina Qayyum	2026-03-03	2026-03-03	In Progress	Do NOT mention TSQLV6 in proposition.
T9	Prepare GitHub repo with full deliverables		Amrina Qayyum	2026-03-03	2026-03-04	Not Started	Commit messages + file naming as required.
T10	Record MP4 walkthrough (planning + SQL demos + statements)	MP4 file saved locally	Amrina Qayyum	2026-03-04	2026-03-04	Not Started	Show To-Do + Gantt, run queries, show FY output, NACE, LLM disclosure.
T11	Upload MP4 to YouTube + submit BrightSpace links	YouTube link + GitHub repo link submitted (leader only)	Amrina Qayyum	2026-03-04	2026-03-04	Not Started	YouTube is the only accepted video format per instructions.

Figure 1: PPG_6 HW4 To-Do List showing tasks, owner, dates, and status.

Gantt Chart (Screenshot)

10 NACE Competencies Used

During completion and presentation, the following NACE Career Readiness Competencies were leveraged:

- **Teamwork:** Coordinated work and reviewed SQL logic collaboratively.
- **Communication:** Explained SQL steps and outputs clearly in the MP4 for non-technical users.
- **Critical Thinking:** Debugged schema/column differences and corrected SQL accordingly.
- **Technology:** Used SQL Server/SSMS (and optionally Azure Data Studio), GitHub, and YouTube.
- **Professionalism:** Followed submission rules, file naming conventions, and deadline planning.
- **Leadership:** Managed task tracking and consolidated deliverables for final submission.

Task ID	Task	Owner	Start	End	Status
T1	Restore databases in SSMS (TSQLV6 + Northwinds2024Student)	Amrina Qayyum	2026-03-02	2026-03-02	Done
T2	Run Chapter 4 (Subqueries) exercises in TSQLV6	Amrina Qayyum	2026-03-02	2026-03-02	Done
T3	Run Chapter 5 (Table Expressions) exercises in TSQLV6	Amrina Qayyum	2026-03-02	2026-03-02	Done
T4	Create FY Quarter scalar function + report in TSQLV6	Amrina Qayyum	2026-03-02	2026-03-02	Done
T5	Map Northwinds2024Student schema/tables/columns	Amrina Qayyum	2026-03-02	2026-03-02	Done
T6	Convert all Chapter 4+5 queries to Northwinds2024Student	Amrina Qayyum	2026-03-02	2026-03-03	Done
T7	Create FY Quarter scalar function + report in Northwinds2024Student	Amrina Qayyum	2026-03-03	2026-03-03	Done
T8	Write Problem Proposition (Northwinds only)	Amrina Qayyum	2026-03-03	2026-03-03	In Progress
T9	Prepare GitHub repo with full deliverables	Amrina Qayyum	2026-03-03	2026-03-04	Not Started
T10	Record MP4 walkthrough (planning + SQL demos + statements)	Amrina Qayyum	2026-03-04	2026-03-04	Not Started
T11	Upload MP4 to YouTube + submit BrightSpace links	Amrina Qayyum	2026-03-04	2026-03-04	Not Started

Figure 2: PPG_6 HW4 Gantt Chart timeline used for planning.

11 LLM / AI Use Statement

We took help from AI for our better understanding and when we faced problems. AI/LLM was used to support learning, debugging, and improving documentation clarity, while ensuring the final SQL solutions were executed and verified in the required databases.

12 GitHub Repository Evidence

A GitHub repository was prepared to store the complete deliverables, including:

- **sql/**
  - **TSQLV6/**
    - `TSQLV6 | Chapter 4 + Chapter 5 + FY Quarter (FULL ANSWERS).sql`
  - **Northwinds2024Student/**
    - `Individual_PPG_6_HW4_AmrinaQayyum_Northwinds.sql`
- **report/**
  - `HW4_Report_AmrinaQayyum.pdf`
  - `todo.png`
  - `gantt.png`

The GitHub repository page was shown in the MP4 to provide proof of project organization and deliverables.

13 Conclusion

This HW4 submission demonstrates correct use of SQL subqueries and table expressions, validated first in TSQLV6 and then executed in Northwinds2024Student as required. The project included a fiscal-year quarter analysis, planning documentation (To-Do and Gantt), a clear business proposition, and a recorded MP4 walkthrough supported by GitHub deliverables.